

Project Proposal

Detachable Tool-Changing Mechanism for Unitree G1 Hand

Course: ELEC4060 Robotics and Automation / COMP5050 Robotics

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Project Group: Group 1

Team Members:

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Submission Date: 29 March 2026

1. Introduction

This project focuses on a detachable mechanism for the Unitree G1 hand that allows rapid interchange between a normal humanoid hand module and three task-specific end-effectors: a pneumatic sucker, a mini drill, and a grapple tool. The selected component is the wrist-side interface and locking mechanism, which is critical because it determines mechanical reliability, safety, and response time during tool switching.

The motivation is to improve robot versatility without redesigning the full arm. Similar to a CNC machine that automatically changes cutting tools for different operations, this robot swaps end-effectors to match each task. Instead of one fixed hand for all tasks, the robot can switch tools based on the operation (pick-and-place with suction, surface operation with drilling, and object capture with grappling). This project applies robotics concepts from the course, including kinematic interface constraints, torque/load transfer, and actuator-control integration for repeatable tool engagement.

The practical outcome is a modular hand platform that reduces downtime, increases task coverage, and provides a scalable foundation for future autonomous tool-changing in humanoid robots.

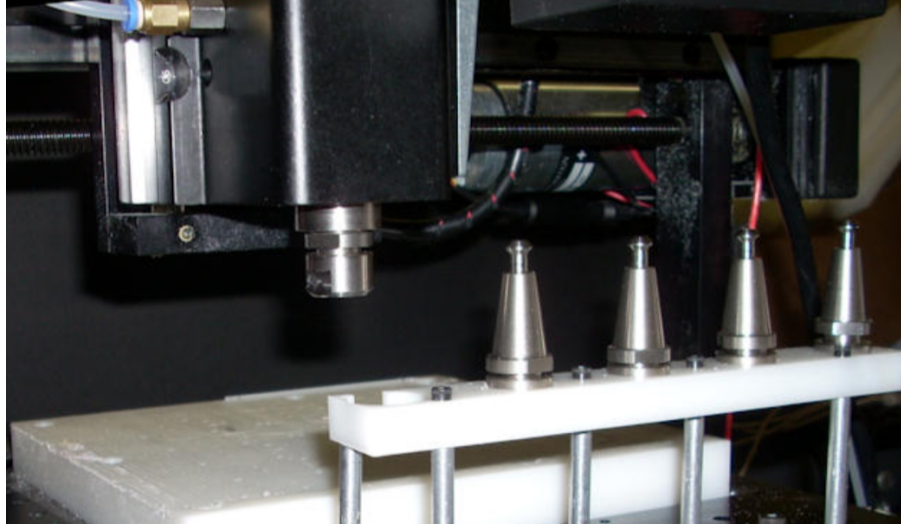


Figure 1: Concept illustration of the detachable tool-changing mechanism.

2. System Description

The proposed system includes a wrist-mounted docking interface on the Unitree G1 side, a fixed tool docking station in the workspace, and four detachable modules: one standard humanoid hand and three tools (pneumatic sucker, drill, and grapple). The docking station holds each module at a known pose for repeatable pickup and return operations. Due to hardware constraints, a simplified Unitree G1-inspired robotic arm will be designed and fabricated to validate the detachable wrist tool-changing mechanism.

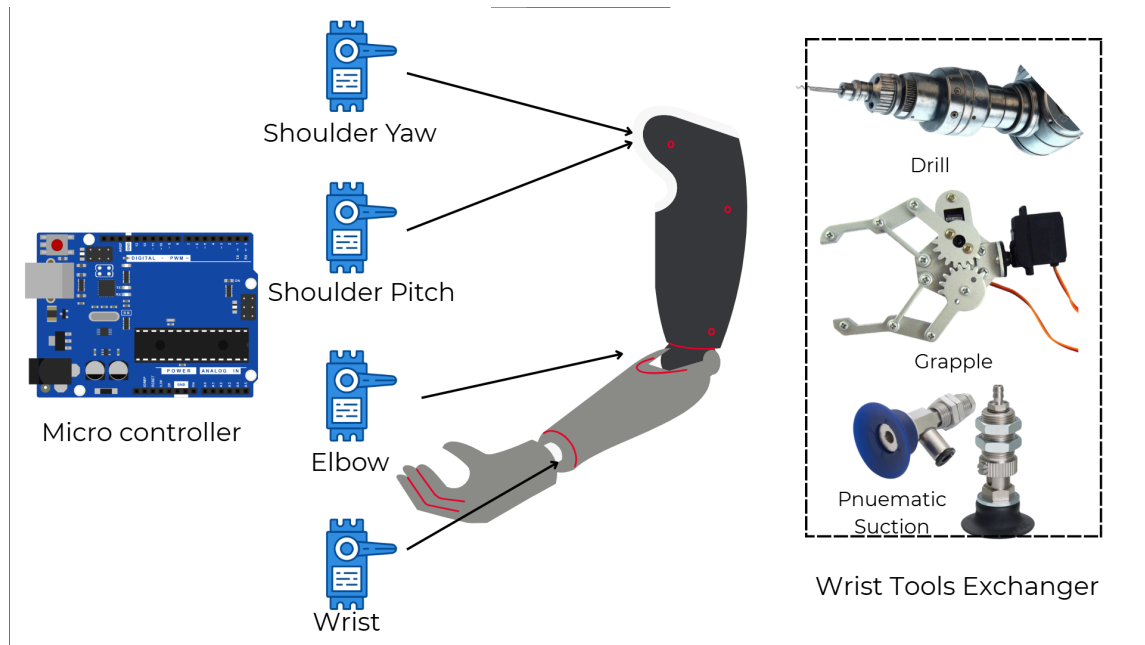


Figure 2: System overview image of the Unitree G1 wrist interface and docking concept.

2.1 Project Overview

The main idea is a quick-swap mechanical and electrical interface that lets the robot change end-effectors in less than 30 seconds. The system automatically recognizes which tool is attached and activates the matching control mode. Because each tool only requires two states (ON/OFF), a single unified control panel is reused for all modules, while labels and safety interlocks update according to detected tool type. The prototype will be developed at an approximately 1:1 wrist-interface scale for realistic fit testing. Expected output includes a functioning lock-release mechanism, stable power/signal connection, automatic tool identification, and validated operation of all three tool modules.

2.2 Design Specifications

- Scale ratio: 1:1 (wrist interface and tool mount)
- Locking actuation: servo-driven or solenoid-driven latch with spring-assisted alignment pins
- Tool actuation: vacuum pump for sucker, DC motor for drill, micro-servo mechanism linkage for grapple
- Materials: PLA/ABS for rigid housing, TPU pads for vibration and grip, steel fasteners, brass inserts, nuts, and bolts
- Power supply: 12 V tool bus with 5 V regulated control rail
- Docking station: fixed multi-slot rack with mechanical guides and passive electrical contacts
- Control method: microcontroller-based switching (ESP32/Arduino) with one shared ON/OFF panel and context-aware labels/interlocks
- Automatic change mode: user selects tool from panel, then robot executes approach-align-lock-verify sequence

2.3 Working Principle

1. The operator (or control script) selects a target tool from the control panel.
2. The robot moves to the fixed docking station coordinates for automatic exchange.
3. The controller verifies that the wrist is in a safe docking posture and disables active tool output.
4. The servo latch unlocks the currently attached module and places it into its assigned docking slot.
5. The robot aligns with the selected slot, picks the selected module using guide pins and a tapered socket, then locks it with the latch.
6. Electrical contacts engage automatically for power and control communication.
7. The controller identifies the attached module and maps the same ON/OFF panel to the correct output channel (vacuum pump, drill motor, or grapple servo).
8. The system performs a quick status check (lock state, current draw, tool response) before enabling operation.

3. Development Timeline with Milestones

- **Week 1 – Define Requirements & Concept**
 - Define major tasks, project scopes, and workspace constraints.
 - Draft sketches and diagrams for the 2D and 3D concept design.
 - Finalize system-level requirements list.
- **Week 2 – Mechanism Design**
 - Design wrist coupler geometry and locking-latch mechanism.
 - Design the humanoid arm.
 - Create 2D designs for the tool end-effectors.
- **Week 3 – Tool Design**
 - Model suction, drill, and grapple end-effector modules.
 - Standardize tool-side interface to match wrist coupler.
 - Route power/control paths for each module.
- **Week 4 – Actuation & Electronics Testing**
 - Test servo/motor ratings on the robotic arm.
 - Test pump and driver modules.
 - Design power architecture (12V bus + 5V regulation).
- **Week 5 – Detailed CAD & Simulation**
 - Complete full assembly CAD with all interfaces.
 - Run kinematic motion checks and interference detection.
 - Perform MATLAB-based trajectory and switching simulations.
 - Conduct optional structural checks on critical parts.
- **Week 6 – Fabrication (Phase 1)**
 - 3D-print coupler, docking fixtures, and tool housings.
 - Post-process parts (drilling, tapping, insert installation).
 - Manufacture or prepare non-printed structural components.
- **Week 7 – Fabrication (Phase 2)**
 - Assemble electronics, drivers, and control board.
 - Complete cable routing and connector termination.
 - Perform bench test for each actuator and tool channel.
- **Week 8 – Integration**
 - Integrate robot arm, coupler, docking station, and all tools.
 - Calibrate coordinate frames for docking and pickup poses.
 - Implement automatic tool-change sequence logic.
- **Week 9 – Testing & Optimization**

- Measure switching time, lock repeatability, and tool response.
- Run reliability cycles and fault-condition checks.
- Tune control parameters for smoother operation.
- Record performance data and update design revisions.

- **Week 10 – Final Validation**

- Execute final demonstration with multiple task scenarios.
- Validate safety behavior and emergency procedures.
- Compile final report, BOM, and test evidence.
- Prepare presentation slides and demo video.

4. Work Distribution Plan

The team consists of three members with complementary engineering backgrounds.

Team Member	Responsibilities
Ngo Hai Khanh (Electrical Engineering)	Power distribution of the system, Control algorithm, System integration and path planning between arm and docking station.
Doan Chi Dung (Mechanical Engineering)	CAD and structural design of the wrist coupler and fixed docking station, tolerance/fit analysis, material selection, fabrication, and assembly validation.
Le Hai Trung (Mechatronic Engineering)	Integration of mechanical and electrical subsystems, primary design and concept testing, 3D design and manufacturing, and final validation.

Table 1: Work distribution among three team members.

5. List of Materials Requested

Materials and components are divided by module for clearer budgeting, procurement, and integration planning.

5.1 Core Platform: Unitree G1 Robotic Arm

In this project, the team will 3D-print, fabricate, and assemble the robotic arm in-house instead of using a fully lab-provided arm.

Item	Qty	Purpose	Unit Cost	Total Cost	Link
3D-printed arm structure	1 set	Fabrication of link structures and mounting interfaces for the custom robotic arm	0	0	
JX PDI-6221MG 20KG digital metal-gear servo	4	Joint actuation for primary arm lift/rotation, reach extension/folding, and wrist end-orientation/tool alignment	229,000 VND	916,000 VND	Link
16-channel PWM controller board (PCA9685)	1	Multi-servo PWM expansion and joint signal control	110,000 VND	110,000 VND	Link
Step-down converter (20A/300W)	1	Voltage conversion for stable logic/control power rails	135,000 VND	135,000 VND	Link
ESP32-S3-WROOM-1 N16R8	1	Main embedded controller for sequencing, communication, and control logic	156,000 VND	156,000 VND	Link
High-torque servo motor (MG90S)	2	Lock/unlock latch actuation and coupling at wrist interface	68,000 VND	136,000 VND	Link

Table 2: Bill of Materials for robotic arm.

5.2 Module A: Pneumatic Sucker

Item	Qty	Purpose	Estimated Cost	Link
Mini vacuum pump + suction cup set	1 set	Suction pickup for flat and lightweight objects	51,000 VND	Link
DC pump 385 (12VDC, 3W, 1.8 L/ph)	1	Vacuum generation for pneumatic suction system	47,000 VND	Link
Silicone pneumatic tube + fittings	1 set	Air path from pump to suction end-effector	15,000 VND	Link

Table 3: Materials for pneumatic sucker module.

5.3 Module B: Mini Drill

Item	Qty	Purpose	Estimated Cost	Link
Mini hand drill set (reused)	1	Reuse an existing mini hand drill set for drilling operations	0	N/A
Protective cover (3D printed)	1	Improves user safety and protects surrounding parts	0	N/A

Table 4: Materials for mini drill module.

5.4 Module C: Grapple Tool

Item	Qty	Purpose	Estimated Cost	Link
Metal gear servo (MG90S)	1	Jaw opening/closing actuation for grasping (same model as latch servo in Table 2)	68,000 VND	Link
Linkage arms + pins (3D printed)	1 set	3D-printed linkage and pin set for synchronized jaw motion	0	N/A
Gripper jaw set (TPU printed)	1 set	TPU-printed compliant jaw contact surfaces	0	N/A
Return spring (already available)	1	Existing spring reused from available stock	0	N/A

Table 5: Materials for grapple module.

6. Expected Outcome

- A functional detachable wrist interface for Unitree G1-compatible hand/tool interchange
- Validated operation of the standard humanoid hand and three tool modules: pneumatic sucker, mini drill, and grapple
- Repeatable interchange process with measured lock stability and switching time
- Comparative task testing results showing strengths of each tool module

7. References

1. Unitree G1 Developer Documentation.
2. Course lecture notes for ELEC4060 / COMP5050.